Karl Kristian Engelund

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Research Interests

Mathematical Finance; Probability Theory; Sequential Statistical Decision Theory; Optimal Stopping; Nonlinear Filtering; Stochastic Control; Numerical Methods.

Publications and Manuscripts

Preprints and Submitted Manuscripts

1. K. K. Engelund. "Regime Detection in Latent Factor Models – A Bayesian Decision-Theoretic Approach." arXiv:2610.20366 (2026). Revised manuscript under review.
2. S. Campbell and K. K. Engelund. "When Should One Stop the Most Exciting Game? Sequential Inference for Win-Martingales." arXiv:2608.12291 (2026). Submitted.
3. S. Campbell and K. K. Engelund. "Embedding Martingale Diffusions as Binary Posteriors in Sequential Inference." arXiv:2607.20373 (2026). Submitted.

Manuscripts in Preparation

1. K. K. Engelund, K. Glover, and H. Hulley. "Caught with Your Shorts Down: Optimal Short Selling with Constraints." *In preparation.*
2. K. K. Engelund, J. B. G. Schreiner, and J. L. Pedersen. "Arbitrary-Precision Upper Approximations of Optimal Exercise Boundaries for American Put Options." *In preparation.*
3. K. K. Engelund. "Sequential Testing of a Doubly Hidden Signal in a Brownian Drift Model." *In preparation.*

Education

University of Copenhagen 2023–present
PhD in Mathematical Finance, Department of Mathematical Sciences
Supervisor: Jesper Lund Pedersen. Co-supervisor: Thomas Mikosch. Expected completion: Summer 2027.

University of Copenhagen 2021–2023
MSc in Actuarial Mathematics
Thesis: *Bayesian Optimal Prediction with Asymmetric Loss.* Thesis grade: 12/12. Supervisor: Jesper Lund Pedersen.

University of Copenhagen 2018–2021
BSc in Mathematics
Thesis: *Stochastic Integration of Continuous Semimartingales.* Thesis grade: 12/12. Supervisor: Ernst Hansen.

Research Visits

Columbia University, USA Feb. 2026–Aug. 2026
Visiting Scholar. Host: Steven Campbell. Research on sequential inference, posterior martingales, and general optimal stopping theory.

University of Technology Sydney, Australia Aug. 2025–Dec. 2025
Visiting Scholar. Host: Kristoffer Glover. Research on optimal short-selling with constraints, margin risk, and applications in market microstructure and digital finance.

Columbia University, USA Jan. 2025–Jul. 2025
Visiting Scholar. Host: Richard Davis. Research on sequential testing for posterior win martingales and quickest detection of volatility regime changes.

Presentations

Conference Presentations and Posters

- *When Should You Stop the Most Exciting Game?*, Talk (forthcoming), *Stochastic Analysis, Statistics of Random Processes and Optimization*, Bakuriani, Georgia, December 2026.
- *When Should You Stop the Most Exciting Game?*, Talk, *9th International Workshop in Sequential Methodologies*, Washington, DC, June 2026.
- *Quickest Detection of Volatility Change in Stochastic Volatility Models*, Talk, *Quantitative Methods in Finance*, Sydney, Australia, December 2025.
- *High-Precision Approximation of the American Put Option*, Poster, *DSI Financial and Business Analytics Poster Session*, Columbia University, New York, NY, January 2025.

Seminar Presentations

- *Embedding and Stopping Exciting Games*, Insurance and Economics Seminar, University of Copenhagen, Copenhagen, Denmark, August 2026.
- *Quickest Change-Point Detection in Stochastic Volatility Models*, Optimal Stopping Seminar, Columbia University, New York, NY, February 2026.
- *Quickest Detection of Volatility Change in the Heston Model*, PhD Workshop, University of Technology Sydney, Sydney, Australia, October 2025.
- *Dimension Reduction Through Canonical Transformations with Applications in Sequential Testing*, Optimal Stopping Seminar, Columbia University, New York, NY, April 2025.
- *Arbitrary-Precision Upper Approximations of Optimal Exercise Boundaries for American Put Options*, Insurance and Economics Seminar, University of Copenhagen, Copenhagen, Denmark, June 2024.
- *Asymmetric Bayesian Optimal Prediction*, Insurance and Economics Seminar, University of Copenhagen, Copenhagen, Denmark, December 2023.

Teaching Experience

- *Finance 1*, Teaching Assistant, University of Copenhagen 2021

- *Optimal Stopping with Applications*, Guest Lecturer and Project Supervisor, University of Copenhagen 2023
- *Mathematical Finance*, Co-designer of Exercises and Assessments, Teaching Assistant, and Guest Lecturer. University of Copenhagen 2024, 2026

Grants

IWSM Funding Apr. 2026
Conference scholarship [$584 USD]

Travel Grant PhD Feb. 2026–Aug. 2026
Travel grant from the University of Copenhagen. [$10,000 USD]

Rejselegat for Matematikere Nov. 2023
Danish travel grant for mathematicians [$94,000 USD].

Professional Experience and Certificates

Certificate in Quantitative Finance, CQF Institute Jan. 2025–Aug. 2025
Professional certificate in quantitative finance, completed part-time. Final project: *Deep Learning for High-Dimensional Asset Pricing*.

QBite Capital Jun. 2022–Nov. 2022
Junior Research Engineer. Quantitative finance research at a small hedge fund focused on digital assets. Developed trading strategies, evaluated the statistical properties of data infrastructure, and implemented quickest-detection methods to improve sampling.

PwC Denmark Nov. 2018–Jun. 2022
Student Assistant. Developed VBA scripts to automate auditing workflows for SME and mid-market clients; worked on data cleaning, fraud detection, and automated reporting.

Technical Skills

Python; R; Bash; C/C++; LaTeX; Monte Carlo methods; numerical PDEs; particle filtering; Codex.

Languages

Danish: native. English: fluent.

Last updated: October 2026